

Furkan Doğan

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PROFESSIONAL SUMMARY

ML Engineer and AI researcher with M.Sc. from Koç University. 3 years of experience building production ML systems. Focus on LLMs, recommendation systems, and MLOps.

EDUCATION

Koç University — M.Sc. Computational Science & Engineering

2019 – 2021 | GPA: 3.90 / 4.00 — Thesis: 'Efficient Transformer Pruning for Edge Deployment'

Koç University — B.Sc. Electrical & Electronics Engineering

2015 – 2019 | GPA: 3.65 / 4.00 — Valedictorian

LANGUAGES

- English: C2 — Proficient (TOEFL iBT 115)
- German: A2 — Elementary

WORK EXPERIENCE

ML Engineer — Şirket.ai — Istanbul (2022 – Present)

Built end-to-end recommendation engine (collaborative filtering + content-based) increasing CTR by 22%. Deployed LLaMA 2 fine-tuned model via vLLM for customer support automation. Managed ML platform on GCP Vertex AI.

AI Research Assistant — Koç University — KUIS AI Lab (2019 – 2022)

Research on neural network pruning and quantization. 3 conference publications (ICML workshop, NeurIPS workshop, ECML).

PROJECTS

- Transformer Pruning Library:** Open-source PyTorch library for structured pruning of BERT/GPT models. 600+ GitHub stars, used by 3 research groups.
- Ollama Model Benchmark Suite:** Automated benchmarking framework for local LLMs. Compares Llama, Mistral, Phi on Turkish NLP tasks.
- Real-time Fraud Detection:** Graph neural network (PyG) for transaction fraud detection. 99.1% precision on holdout set.

INTERNSHIPS

AI Research Intern — Microsoft — Redmond, WA (Remote) (Summer 2020)

Contributed to Azure Cognitive Services NLP evaluation pipeline.

Data Science Intern — Garanti BBVA Technology (Summer 2018)

Churn prediction model for retail banking using LightGBM.

TECHNICAL SKILLS

Python, PyTorch, TensorFlow, HuggingFace, vLLM, Ollama, FastAPI, GCP, Docker, MLflow, DVC, Kafka, SQL

AI & MACHINE LEARNING

Expert-level AI: 3 ML research publications. Fine-tuned LLaMA 2 and Mistral for production. Deep expertise in transformer architecture, pruning, quantization (GGUF, GPTQ, AWQ). Built and deployed RAG pipelines. Familiar with LangChain, LlamaIndex, ChromaDB, Weaviate.